

MADHAVAN S

FULL-STACK DEVELOPER | AI APPLICATION BUILDER

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SUMMARY

Full-stack developer specializing in secure, concurrent backend systems and applied AI integrations, built primarily with Python/Django and Next.js/TypeScript. Track record of auditing and remediating real production vulnerabilities (CVE-level RCE fixes), implementing row-level locking for race-condition-free transactions, and shipping deployed, end-to-end applications rather than tutorials.

EDUCATION

B.Sc. Computer Science

2023 – 2026

KMG College of Arts & Science (Thiruvalluvar University), Vellore

TECHNICAL SKILLS

- **Languages & Backend:** TypeScript, Python, JavaScript (ES6+), Next.js (App Router), Django REST Framework, Flask, REST APIs, Celery, Redis
- **Data & Cloud:** Supabase (Postgres, Auth, Storage, RLS), Neon PostgreSQL, Firebase, SQLite, GitHub Actions CI
- **Frontend:** React 19, Tailwind CSS, Radix UI / shadcn, React Hook Form + Zod, PWA
- **AI & Vision:** Google Gemini, Google Genkit, DeepFace, OpenCV, Prompt Engineering
- **Testing & Security:** Vitest, React Testing Library, Django TransactionTestCase, CVE Remediation, Concurrency Control, Cryptographic Auth, Row-Level Security

WORK EXPERIENCE

Nextgen Technology Pvt. Ltd. | Python Developer Intern

May 2025 – Jun 2025

Vellore, India

- Built a Django-based bus reservation system with SQLite, implementing complete session flows from route search through booking confirmation for a multi-stage passenger workflow.
- Designed a relational schema via Django ORM to map bus schedules, seat inventory, and passenger records, supporting accurate real-time seat availability tracking.
- Implemented server-side form validation and CSRF protection across all booking endpoints, closing edge cases including duplicate submissions and invalid seat selections.

ENGINEERING PROJECTS

Vivasayi — AI Farm Management Platform

2025 – Present

Live: vivasayi-wheat.vercel.app | Tech: Next.js 15, TypeScript, React 19, Supabase, Google Gemini via Genkit, Vitest

- Built a production-deployed farm management platform with 4 distinct AI-backed tools (crop diagnosis, advisory, and record-keeping) for Indian farmers, securing all user data with Supabase row-level security.
- Identified and resolved a critical Next.js remote code execution vulnerability (GHSA-9qr9-h5gf-34mp) and a jsPDF dependency CVE chain through a full security audit, taking the application from multiple critical CVEs to zero.
- Engineered AI call resilience with exponential backoff, automatic fallback to a secondary Gemini model, and quota-type detection to distinguish transient rate limits from daily caps — preventing user-facing failures during API exhaustion.
- Architected a 23-language internationalization system (React context + custom hook) to support India's linguistic diversity, with active translation coverage across major languages including Tamil and Hindi.

Havan Bus Booking Engine — High-Concurrency System

2025

Live: havan-bus-booking-engine.vercel.app | Tech: Django REST Framework, Celery, Redis, Neon PostgreSQL, React, GitHub Actions (scheduled jobs)

- Built a decoupled, production-style bus reservation REST API using Django's `select_for_update()` row-level locking to serialize concurrent seat-booking transactions at the database level.
- Validated the locking strategy with a multithreaded test firing 50 simultaneous booking requests at the same seat, confirming exactly one transaction succeeded and zero double-bookings occurred.
- Closed a timing-attack vector in internal token comparison by switching to `hmac.compare_digest()` for constant-time evaluation.
- Offloaded synchronous PDF ticket generation to an asynchronous Celery + Redis pipeline, decoupling slow report rendering from the request/response cycle to keep the booking API responsive.

CERTIFICATIONS & LANGUAGES

- **Certifications:** Python & OOP, Data Science & Analytics, Agile Methodology (Infosys Springboard)
- **Languages:** Tamil (Native), English (Professional)